1341C Test 4 v1

1. If G is an $n \times 2$ matrix and $H = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ then the second column of the matrix GH is

A. not defined unless $n = 2$.
(B.) the same as the second column of G.
C. the same as the second column of H.
D. the same as the first column of G.
E. the same as the first column of H.
F. the sum of the first and the second column of G.

Let $G = [C_1 \; C_2]$ (block column form). Then $GH = [C_1 \; C_2] \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

$= [C_1 + C_2 \quad C_2]$ (block column form)

Hence B is correct

2. If $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$, then the third row of B^{-1} is:

A. [0 1 0]
B. [1 0 2]
(C.) [−1 1 1]
D. [0 0 1]
E. [1 0 −1]
F. [1 0 1]

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{array}\right]$$ Further

reduction will not change the 3rd row of . Hence

C is correct

3. What is the dimension of the subspace of $\mathbf{R}^5$ spanned by $(1,-1,4,-5,0)$, $(2,1,5,-1,0)$, and $(0,1,-1,3,0)$?

A. 0
B. 1
C. 2
D. 3
E. 4
F. 5

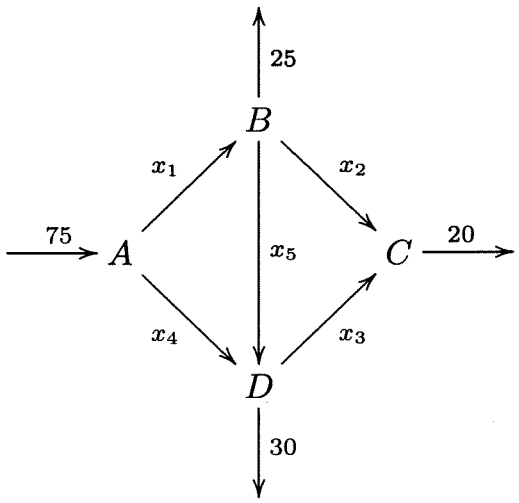
This will be rank $\begin{bmatrix} 1 & -1 & 4 & -5 & 0 \\ 2 & 1 & 5 & -1 & 0 \\ 0 & 1 & -1 & 3 & 0 \end{bmatrix}$

$= \text{rank} \begin{bmatrix} 1 & -1 & 4 & -5 & 0 \\ 0 & 3 & -3 & 9 & 0 \\ 0 & 1 & -1 & 3 & 0 \end{bmatrix}$

$= \text{rank} \begin{bmatrix} 1 & -1 & 4 & -5 & 0 \\ 0 & 1 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$= 2$

4. Consider the network of streets with intersections A, B, C and D below. The arrows indicate the direction of traffic flow along the one way streets, and the numbers refer to the number of cars observed to enter A or leave B, C and D during one minute. Each x_i denotes the unknown number of cars which passed along the indicated streets during the same period.



a) Write down the linear system which describes the the traffic flow, **together with all the constraints** on the variables x_i, $i = 1, \ldots, 5$. (Do not perform any operations on your equations: this is done for you in (b).)

Intersection — Flow in = Flow out

A: $75 = x_1 + x_4$

B: $x_1 = x_2 + x_5 + 25$

C: $x_2 + x_3 = 20$

D: $x_4 + x_5 = x_3 + 30$

Constraints: $x_i \geq 0 \quad i = 1, \ldots, 5$

$x_i \in \mathbb{Z}, \quad i = 1, \ldots, 5$

5b) The reduced row-echelon form of the augmented matrix from part (a) is

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & | & 75 \\ 0 & 1 & 0 & 1 & 1 & | & 50 \\ 0 & 0 & 1 & -1 & -1 & | & -30 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}$$

(s, t)

Find the general solution. (Ignore the constraints at this point.)

$x_1 = 75 - s$
$x_2 = 50 - s - t$
$x_3 = -30 + s + t$; $s, t \in \mathbb{R}$
$x_4 = s$
$x_5 = t$

c) If BD were closed due to roadwork, find the minimum flow on AB, using your results from (b).

• BD closed $\Leftrightarrow x_5 = 0 \Leftrightarrow t = 0$. Find min $x_1 = 75 - s$

Constraints

$x_1 \geq 0 \Leftrightarrow s \leq 75$
$x_2 \geq 0 \Leftrightarrow s \leq 50$
$x_3 \geq 0 \Leftrightarrow 30 \leq s$
$x_4 \geq 0 \Leftrightarrow 0 \leq s$

$\Rightarrow 30 \leq s \leq 50$.

Hence the smallest value of $x_1 = 75 - 50 = 25$

5. Let $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 5 & 10 & -4 & 19 \\ 2 & 4 & -2 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ (*)

a) Find a basis for the row space row (A) of A.

b) Find a basis for the column space col (A) of A.

c) Find the dimension of $\{Ax \mid x \in \mathbf{R}^4\}$.

d) Find a basis for $\ker(A) = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

i) From (*) and the row space algorithm, $\{(1,2,0,3), (0,0,1,-1)\}$ is a basis for row(A)

ii) From (*) and the column space algorithm, $\{(1,5,2), (-1,-4,-2)\}$ is a basis for col(A).

c) The set $\{Ax \mid x \in \mathbb{R}^4\} = \text{col}(A)$, so by (b), its dimension is 2.

d) $\left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ $\therefore$ $x_1 = -2s - 3t$, $x_2 = s$, $x_3 = t$, $x_4 = t$; $s, t \in \mathbb{R}$ $\therefore$

$\ker A = \text{span}\{(-2,1,0,0), (-3,0,1,1)\}$ and $\{(-2,1,0,0), (-3,0,1,1)\}$ is a basis for $\ker A$

6. (a) State whether each of the following statements is (always) true, or is (possibly) false, in the box after the statement.

- If you say the statement may be false, you must give an explicit example - with numbers!
- If you say the statement is always true, you must give a clear explanation.

i) If A is a 3 by 4 matrix, then the rows of A must be linearly dependent.

eg $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. the rows of A are l.i.

ANSWER FALSE

ii) If A is a 2 by 2 matrix and $A^2 = 0$, then $A = 0$.

Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ Then $A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

but $A \neq 0$.

ANSWER FALSE

6 (b). Let A be a real $n \times n$ matrix. Give 3 statements (in total) equivalent to

"A is invertible",

one each in terms of:

(I) the columns of A

- They are li.
- They span $\mathbb{R}^n$
- They are a basis of $\mathbb{R}^n$

(II) the reduced row-echelon form of A is I_n, the $n \times n$ identity matrix

(III) the homogeneous system $Ax = 0$.

- Has a unique sol^n, $x = 0$

7. [Bonus] Prove carefully that if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfies $ad - bc = 0$, then rank $A < 2$. (Note: you cannot use the formula for the inverse of a 2 by 2 matrix, since the formula relies on $ad - bc \neq 0$.)

Suppose $ad = bc$, but A is invertible.

Now note that if $B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, then

$$AB = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (*)$$

Since A^{-1} is assumed to exist, $*$ implies

$$A^{-1}(AB) = A^{-1}(0) = 0$$

i.e. $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. But then

$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, which is not invertible, a contradiction

Hence A is not invertible.